

3.2

a) znormujeme 1: $\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{2}$

ortogonalizujemo x: $\vec{v}_2 - \vec{w}_2 = \vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1$; $\vec{w}_2 = \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1 = \int_1^5 \frac{1}{2} x dx \cdot \frac{1}{2} = \left(\frac{25}{4} - \frac{1}{4} \right) \cdot \frac{1}{2} = 3$

znormujeme $\vec{w}_2$: $\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \frac{(x-3)\sqrt{3}}{4}$

ortogonalizujemo x^2 : $\vec{w}_3 = \langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2 = \int_1^5 \frac{1}{2} x^2 dx \cdot \frac{1}{2} + \int_1^5 \frac{(x-3)\sqrt{3}}{4} x^2 dx \cdot \frac{(x-3)\sqrt{3}}{4} =$
 $= \frac{31}{3} + 6(x-3) = 6x - \frac{23}{3}$

znormujeme $\vec{w}_3$: $\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\|x^2 - 6x + \frac{23}{3}\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{256}{45}}}$

$$B = \left(\frac{1}{2}, \frac{(x-3)\sqrt{3}}{4}, \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{256}{45}}} \right)$$

b) $\frac{\int_1^5 (x^2 - 6x + \frac{23}{3}) \sin x dx}{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx} (x^2 - 6x + \frac{23}{3}) + \frac{\int_1^5 (x-3) \sin x dx}{\int_1^5 (x-3)^2 dx} (x-3) + \frac{\int_1^5 \sin x dx}{\int_1^5 1^2 dx} \cdot 1 =$

$$\approx -0,053 x^2 - 0,332 x + 1,553$$

$$a = -0,053 \quad b = -0,332 \quad c = 1,553$$

c) $\|\sin x - (-0,053 x^2 - 0,332 x + 1,553)\| = \sqrt{\int_1^5 (\sin x - (ax^2 + bx + c))^2 dx} \approx 0,518$